

| Question | Answer                                                                                                        | Mark                                              | Mark scheme                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Additional guidance                                                                                                                                                                                 |
|----------|---------------------------------------------------------------------------------------------------------------|---------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 23       | Sketch graph with minimum point at $(3p, -9p^2 - 7)$ in 4 <sup>th</sup> quadrant and y intercept at $(0, -7)$ | <p>B1</p> <p>M1</p> <p>M1</p> <p>B1</p> <p>C1</p> | <p>for a curve drawn with intercept at the point <math>(0, -7)</math></p> <p>for the start of a method to find the turning point,<br/>eg <math>(x - 3p)^2 \dots</math><br/><b>or</b> an x coordinate of <math>3p</math> at the vertex</p> <p>(dep) for a complete correct method to find the turning point,<br/>eg <math>(x - 3p)^2 - 9p^2 - 7</math><br/><b>or</b> substitutes <math>x = 3p</math> into the equation to find y</p> <p>for turning point at <math>(3p, -9p^2 - 7)</math></p> <p>for a sketch of a parabola drawn with intercept at <math>(0, -7)</math> and minimum turning point clearly shown as being at <math>(3p, -9p^2 - 7)</math> in the 4th quadrant.</p> | <p>where ... can be any constant term(s) but not term(s) in x</p> <p>Allow <math>(3p)^2</math> for <math>9p^2</math> throughout</p> <p>A correct answer with no supportive working gets 0 marks</p> |